God's Eye View: from audit to compliance

CrossCheck audited God's Eye View against two global standards: **EUROCAE ED-153** for software safety assurance at SWAL 3, and **OWASP ASVS 5.0** for application security at Level 2. This plan closes every gap it found, in four waves, with each fix evidenced line by line.

1	2	3	4	5	6	7	8	9
Baseline frozen	**Audit**	**Plan issued**	**Plan approved**	**Wave 1 · Harden**	**Wave 2 · Control access**	**Wave 3 · Prove safety**	**Wave 4 · Assure and hand over**	**Evidence pack issued**
2026-10-14	2026-10-14	2026-10-14	next	planned	planned	planned	planned	planned

WHERE WE START · BASELINE 1518EA8



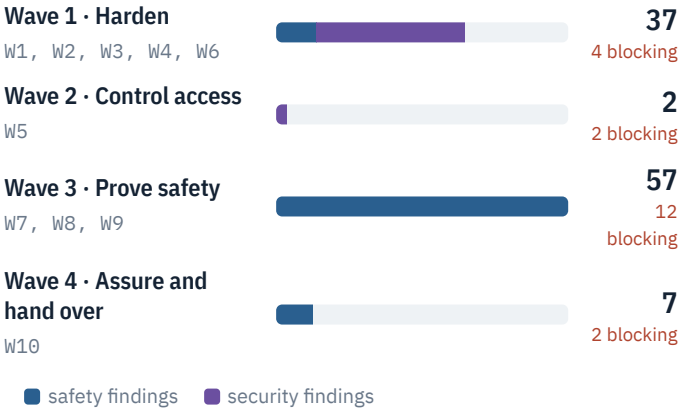
Safety · ED-153
SWAL 3 claim not supported

Security · ASVS L2
ASVS L2 not met

103 findings: 72 safety, 31 security. **20** of them block a claim.

HOW THE GAPS ARE CLOSED

103 findings → 10 workstreams → 4 waves



WHERE WE FINISH

1. **After Wave 2:** ASVS Level 2 met.
2. **After Wave 3:** no S1–S2 safety finding open.
3. **After Wave 4:** ED-153 SWAL 3 claim supported, safety folder issued, ready for independent assessment.

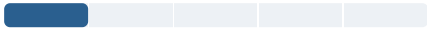
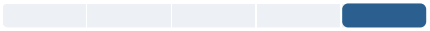
Every fix is a named commit with a test or a versioned document. CrossCheck re-audits after each wave against the frozen baseline.

READING THIS RECORD

0 This page · 1 Waves · 2 Workstreams · 3 Line by line, every finding · A Method and rules

Four waves, each ending in a re-audit

Waves 1–2 close the security gaps in code. Waves 3–4 build the safety evidence that ED-153 asks for. Wave 3 can start alongside Wave 2 if a safety lead is free.

WAVE	WORKSTREAMS	WEEKS 1 → 5	EXIT CHECK
Wave 1 · Harden Engineering · 1 week	W1 Web hardening W2 Abuse resistance and availability W3 Supply chain and secrets W4 Logging and monitoring W6 Data and validation rules		Every Wave 1 security finding closed except V8; re-audit
Wave 2 · Control access Engineering + customer identity provider · 1 week	W5 Access control		ASVS Level 2 met; re-audit
Wave 3 · Prove safety Safety lead, engineering, customer · 2–3 weeks	W7 Safety analysis W8 Safety plan, process and configuration management W9 Requirements, verification and validation		No S1–S2 finding open
Wave 4 · Assure and hand over Safety lead, independent reviewer · 1 week	W10 Safety case and release evidence		SWAL 3 claim supported; evidence pack issued

Only the customer or owner can do these

ACTION	NEEDED BY
Choose the identity provider	Wave 2
Confirm the operational context; join the hazard assessment	Wave 3
Name the plan approver and an independent reviewer	Wave 3
Restrict browser keys in the Google and Cesium consoles	Wave 1
Validate with operational users	Wave 3

Not ours to give: final ED-153 acceptance needs an independent assessor, the customer's operational context, and a check of CrossCheck's reconstructed catalogue against a purchased copy of ED-153.

Ten workstreams; each finding belongs to exactly one

W1 Web hardening

Wave 1

0 safety 6 security

1 blocking

- Security headers on every response: HSTS, nosniff, Referrer-Policy and a Content-Security-Policy with script-src.
- Inline HTML templates moved so the policy can drop unsafe-inline.
- Debug handle (window.__godsEyeView, ?fps) gated out of production builds.

Evidence: Header tests in the preview-serving suite; a browser-captured policy report.

W2 Abuse resistance and availability

Wave 1

1 safety 5 security

1 blocking

- Rate limiting on by default for every cost-bearing proxy, with global caps.
- Client IP taken from the forwarded header only when the request comes through the known proxy hop.
- Heavy endpoints and resource budgets (frame time, memory, latency) documented.

Evidence: Limiter tests; a load test against the deployed app.

W3 Supply chain and secrets

Wave 1

4 safety 6 security

1 blocking

- npm audit and Dependabot in CI; CI fails on High or Critical.
- CycloneDX software bill of materials (SBOM) for each release; patch time limits in SECURITY.md.
- COTS dossiers: CesiumJS, satellite.js, Node, browsers, Replit, each data feed.

Evidence: CI logs; SBOM file; dossiers; console screenshots.

W4 Logging and monitoring

Wave 1

3 safety 8 security

1 blocking

- Structured JSON logs with UTC timestamps and request IDs; untrusted values encoded.
- Security events logged: rate-limit hits, validation rejects.
- Log inventory; logs forwarded to a separate store with alerting (Wave 2).

Evidence: Logger tests; log inventory; an alert fired in test.

W5 Access control

Wave 2

0 safety 2 security

2 blocking

- Sign-in through an external identity provider (OIDC); roles viewer, operator, admin.
- Server-side checks on every function; paid proxies need operator; Exercise mode needs an explicit role.
- Session timeout and logout; authorization model documented.

Evidence: Access-control test matrix (role × endpoint); CrossCheck assessment of the new V6, V7, V9 and V10 rows.

W6 Data and validation rules

Wave 1

0 safety 4 security

0 blocking

- Input-validation specification per API endpoint: types, ranges, lengths, combined-field rules.
- Data classification (public, aggregated-sensitive, simulated) with handling rules for caching, export and retention.

Evidence: Both documents, each validation rule traced to an existing test.

W7 Safety analysis

Wave 3

10 safety 0 security

8 blocking

- Concept of operations: users, environment, advisory-only boundary.
- Functional hazard assessment: failure modes per component, effects on the operator, justified likelihood.
- SWAL allocation per failure; preliminary system safety assessment.

Evidence: Versioned analysis documents reviewed with the customer.

W8 Safety plan, process and configuration management

Wave 3

29 safety 0 security

3 blocking

- Software safety assessment plan: approach, rigour per SWAL, I/R/D classification, named reviewer sign-off.
- Development plan with an AI-assistance policy: no tool credit, human review of every AI change.
- Configuration management plan: protected main, reviews, tagged releases, CI-built artifact with build stamp, tested rollback.

Evidence: Approved plans; repository settings evidence; release records.

W9 Requirements, verification and validation

Wave 3

18 safety 0 security

1 blocking

- Reviewed and approved SRS covering the inherited layer engine; traceability widened from 56 requirements.
- Dead-code review (remove /api/route); isolation argument per first-level component.
- Integration tests at every component boundary; feed-integrity and outage tests.

Evidence: Review records, test reports, validation minutes.

W10 Safety case and release evidence

Wave 4

7 safety 0 security

2 blocking

- Software safety assessment report with a GSN-style argument (direct and back-up).
- Software safety folder; assessment report and distribution record per release; independent audit.

Evidence: Issued safety folder and independent audit report.

Every finding, its fix and how it will be closed

"Closed by" is filled from the ledger as fixes land: a commit SHA, test or document reference, with the date and new rating in CrossCheck.

W1 Web hardening

Wave 1 · 6 findings

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
V3.2.1 ASVS	Security controls are in place to prevent browsers from rendering content or functionality in HTTP responses in an incorrect context (e.g., when an... V3 Web Frontend Security	Partial	Medium	Add nosniff and a CSP (see V3.4.3, V3.4.4).	API responses set Content-Type; no X-Content-Type-Options or CSP sandboxing.	open
V3.4.1 ASVS	A Strict-Transport-Security header field is included on all responses to enforce an HTTP Strict Transport Security (HSTS) policy. A maximum age of at... V3 Web Frontend Security	Non-compliant	Medium	Set HSTS (max-age >= 1 year) in server headers or at the edge.	No Strict-Transport-Security header set by the app; edge behaviour on Replit unverified.	open
V3.4.3 ASVS	HTTP responses include a Content-Security-Policy response header field which defines directives to ensure the browser only loads and executes trusted... V3 Web Frontend Security	Non-compliant	High	Ship a script-src CSP (nonces or hashes) after moving inline HTML templates.	CSP is frame-ancestors only; no script-src/object-src/base-uri (SECURITY.md known gap 2).	open
V3.4.4 ASVS	All HTTP responses contain an 'X-Content-Type-Options: nosniff' header field. This instructs browsers not to use content sniffing and MIME type... V3 Web Frontend Security	Non-compliant	Medium	Add nosniff to server.headers.	X-Content-Type-Options: nosniff not set.	open
V3.4.5 ASVS	The application sets a referrer policy to prevent leakage of technically sensitive data to third-party services via the 'Referer' HTTP request header... V3 Web Frontend Security	Partial	Low	Set Referrer-Policy: strict-origin-when-cross-origin explicitly.	No explicit Referrer-Policy; browser default strict-origin-when-cross-origin applies.	open
V13.4.5 ASVS	Documentation (such as for internal APIs) and monitoring endpoints are not exposed unless explicitly intended. V13 Configuration	Partial	Low	Remove or gate the debug handle in production.	window.__godsEyeView debug handle exposed in production builds.	open

W2 Abuse resistance and availability

Wave 1 · 6 findings

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
V2.4.1 ASVS	Anti-automation controls are in place to protect against excessive calls to application functions that could lead to data exfiltration, garbage-data... V2 Validation and Business Logic	Partial	High	Enable limits by default; add global quota caps and alerting on provider spend.	Opt-in per-IP limiter (default unlimited); cost-bearing Google proxies callable by anyone with the URL.	open
V2.3.2 ASVS	Business logic limits are implemented per the application's documentation to avoid business logic flaws being exploited. V2 Validation and Business Logic	Partial	Medium	Turn rate limiting on by default in production config.	Limits exist but per-IP rate limiting is opt-in and off by default.	open
V15.3.4 ASVS	All proxying and middleware components transfer the user's original IP address correctly using trusted data fields that cannot be manipulated by the... V15 Secure Coding and Architecture	Partial	Medium	Trust the platform's forwarded header only from the known proxy hop.	Limiter ignores X-Forwarded-For, so behind the Replit proxy every client may share one bucket.	open
V2.1.3 ASVS	Expectations for business logic limits and validations are documented, including both per-user and globally across the application. V2 Validation and Business Logic	Partial	Medium	Document per-client and global limits for each cost-bearing proxy.	Rate limits and provider budgets described in CONFIGURATION-INDEX; no per-client/global limit policy.	open
V15.1.3 ASVS	The application documentation identifies functionality which is time-consuming or resource-demanding. This must include how to prevent a loss of... V15 Secure Coding and Architecture	Partial	Low	Document heavy endpoints (Overpass, routes, tracks) and their limits.	KNOWN-LIMITATIONS and provider budgets cover upstream cost; no list of resource-heavy endpoints.	open
4.3.13 ED-153	Resource management (load, memory, time response) 4.3 Development	Partial	S4	Resource budgets (frame time, memory, latency).	Benchmark, adaptive quality ladder, caps.	open

W3 Supply chain and secrets

Wave 1 · 10 findings

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
V15.2.1 ASVS	The application only contains components which have not breached the documented update and remediation time frames. V15 Secure Coding and Architecture	Non-compliant	High	Add npm audit and Dependabot; fail CI on high/critical.	No dependency vulnerability scanning in CI (SECURITY.md known gap 4); state cannot be shown.	open

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
V15.1.1 ASVS	Application documentation defines risk based remediation time frames for 3rd party component versions with vulnerabilities and for updating libraries... V15 Secure Coding and Architecture	Non-compliant	Medium	Add a patch SLA to SECURITY.md (e.g. critical 7 days, high 30).	No documented remediation time frames for vulnerable components.	open
V15.1.2 ASVS	An inventory catalog, such as software bill of materials (SBOM), is maintained of all third-party libraries in use, including verifying that... V15 Secure Coding and Architecture	Non-compliant	Medium	Generate a CycloneDX SBOM in CI and keep it with each release.	No SBOM; package-lock.json only.	open
V13.3.1 ASVS	A secrets management solution, such as a key vault, is used to securely create, store, control access to, and destroy backend secrets. These could... V13 Configuration	Partial	Medium	Move to a managed secret store with rotation for the production deployment.	Secrets in Replit Secrets/env; .env.example names only; no vault, rotation or expiry.	open
V13.3.2 ASVS	Access to secret assets adheres to the principle of least privilege. V13 Configuration	Partial	Medium	Restrict browser keys by HTTP referrer and API; verify in the Google and Cesium consoles.	Server keys stay server-side, but GOOGLE_MAPS_API_KEY and CESIUM_ION_TOKEN are bundled to the browser by design.	open
V13.2.5 ASVS	The web or application server is configured with an allowlist of resources or systems to which the server can send requests or load data or files from. V13 Configuration	Partial	Low	Restrict outbound egress to the documented hosts where hosting allows.	No egress restriction at the server or platform level.	open
3.0.13 ED-153	COTS 3 SSAS	Partial	S3	COTS dossiers: CesiumJS, satellite.js, Node, browser, host, feeds.	Licence inventory and data-source register; no COTS dossiers.	open
7.x COTS ED-153	COTS 7 Additional ANS	Partial	S4	COTS dossiers, SBOM, wrappers.	Licence inventory only.	open
7.x SE ED-153	Service experience 7 Additional ANS	Non-compliant	S3	Service history argument where COTS credit claimed.	No service history.	open
4.3.10 ED-153	Standards/rules (incl. PDS/COTS references) 4.3 Development	Partial	S4	Reference PDS/COTS standards and deviations.	Inherited code follows upstream conventions.	open

W4 Logging and monitoring

Wave 1 · 11 findings

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
V16.1.1 ASVS	An inventory exists documenting the logging performed at each layer of the application's technology stack, what events are being logged, log formats,... V16 Security Logging and Error Handling	Non-compliant	Medium	Document what each layer logs, where, and retention.	No log inventory.	open
V16.3.3 ASVS	The application logs the security events that are defined in the documentation and also logs attempts to bypass the security controls, such as input... V16 Security Logging and Error Handling	Non-compliant	Medium	Log rate-limit and validation rejects as security events.	No security events defined or logged (rate-limit hits, validation rejects).	open
V16.4.3 ASVS	Logs are securely transmitted to a logically separate system for analysis, detection, alerting, and escalation. The aim is to ensure that if the... V16 Security Logging and Error Handling	Non-compliant	Medium	Forward logs to a SIEM or log service with alerting.	No log shipping to a separate system.	open
V16.2.1 ASVS	Each log entry includes necessary metadata (such as when, where, who, what) that would allow for a detailed investigation of the timeline when an... V16 Security Logging and Error Handling	Partial	Low	Adopt structured JSON logs with timestamp and request id.	console.warn/error lines carry provider and reason, but no structured when/where metadata.	open
V16.2.2 ASVS	Time sources for all logging components are synchronized, and that timestamps in security event metadata use UTC or include an explicit time zone... V16 Security Logging and Error Handling	Partial	Low	Emit ISO-8601 UTC timestamps.	Relies on platform timestamps; no explicit UTC in log lines.	open
V16.2.4 ASVS	Logs can be read and correlated by the log processor that is in use, preferably by using a common logging format. V16 Security Logging and Error Handling	Partial	Low	Use JSON lines.	Free-text console logs; no common format.	open
V16.4.1 ASVS	All logging components appropriately encode data to prevent log injection. V16 Security Logging and Error Handling	Partial	Low	Encode or JSON-stringify untrusted values in logs.	Some feed-derived strings logged without encoding.	open
V16.4.2 ASVS	Logs are protected from unauthorized access and cannot be modified. V16 Security Logging and Error Handling	Partial	Low	Ship logs to an append-only store.	Logs protected only by platform access control; not tamper-evident.	open

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
3.0.11 ED-153	SWAL monitoring 3 SSAS	Non-compliant	S2	Telemetry and incident log mapped to SWAL.	No in-service malfunction monitoring.	open
4.4.5 ED-153	Performance monitoring 4.4 Operation	Non-compliant	S3	Telemetry and feed-health indicators.	No performance or feed-health monitoring.	open
4.4.4 ED-153	Software operation 4.4 Operation	Partial	S4	Operating procedures and logs.	Server logs only.	open

W5 Access control

Wave 2 · 2 findings

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
V8.1.1 ASVS	Authorization documentation defines rules for restricting function-level and data-specific access based on consumer permissions and resource attributes. V8 Authorization	Non-compliant	High	Define roles (viewer, operator, admin) and which functions each may use.	No authorization model documented; the app is public-by-design (SECURITY.md known gap 1).	open
V8.2.1 ASVS	The application ensures that function-level access is restricted to consumers with explicit permissions. V8 Authorization	Non-compliant	High	Put authentication (SSO/OIDC) in front of the app before any non-public deployment.	Every function, including cost-bearing API proxies and Exercise mode, is open to anyone with the URL.	open

W6 Data and validation rules

Wave 1 · 4 findings

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
V2.1.1 ASVS	The application's documentation defines input validation rules for how to check the validity of data items against an expected structure. This could... V2 Validation and Business Logic	Partial	Medium	Write an input-validation spec per API endpoint (types, ranges, lengths).	Validation exists in code (400 on bad hex, callsign, coordinates, cyclone query) but rules are not documented.	open
V2.1.2 ASVS	The application's documentation defines how to validate the logical and contextual consistency of combined data items, such as checking that suburb... V2 Validation and Business Logic	Partial	Low	Add to the validation spec.	Combined-field checks exist in code (bbox, route coords); not documented.	open
V14.1.1 ASVS	All sensitive data created and processed by the application has been identified and classified into protection levels. This includes data that is... V14 Data Protection	Non-compliant	Medium	Classify data (public, aggregated-sensitive, simulated) and record handling rules.	No data classification. Public OSINT, but aggregated military tracks may be sensitive to a defence customer.	open

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
V14.1.2 ASVS	All sensitive data protection levels have a documented set of protection requirements. This must include (but not be limited to) requirements related... V14 Data Protection	Non-compliant	Medium	Define per-class requirements (retention, caching, export).	No protection requirements per level.	open

W7 Safety analysis

Wave 3 · 10 findings

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
3.1.2 ED-153	Operational environment 3.1 Initiation	Partial	S3	ConOps: users, environment, advisory-only boundary.	Demo use described; no operational concept.	open
3.1.5 ED-153	System FHA & PSSA output 3.1 Initiation	Non-compliant	S2	FHA with the customer.	No FHA/PSSA.	open
4.1.2 ED-153	Functional Hazard Assessment 4.1 Acquisition	Non-compliant	S2	FHA (shared with 3.1.5).	No FHA.	open
3.3.1 ED-153	Failures identification 3.3 Requirements	Non-compliant	S2	Failure modes per component.	No software failure-mode list.	open
3.3.2 ED-153	Failure effects 3.3 Requirements	Non-compliant	S2	Effects of each failure on users.	No effects analysis.	open
3.3.3 ED-153	Assessment of risk 3.3 Requirements	Non-compliant	S2	Likelihood justification.	No risk assessment.	open
3.0.5 ED-153	SWAL allocation 3 SSAS	Non-compliant	S2	Hazard log and severity x likelihood allocation.	SWAL set top-down; no hazard analysis.	open
3.3.5 ED-153	SWAL allocation 3.3 Requirements	Non-compliant	S2	Allocation per failure.	No allocation record.	open
4.1.3 ED-153	Preliminary System Safety Assessment 4.1 Acquisition	Non-compliant	S2	PSSA after the FHA.	No PSSA.	open
3.3.4 ED-153	Software requirements setting 3.3 Requirements	Partial	S3	GEV-SAF tier traced into the SRS.	Functional requirements exist; no safety requirements.	open

W8 Safety plan, process and configuration management

Wave 3 · 29 findings

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
3.0.1 ED-153	Implementation of the SSAS 3 SSAS	Non-compliant	S2	Write the SSAS into a software safety assessment plan.	No documented software safety assurance system.	open
3.2.1 ED-153	Approach 3.2 Planning	Partial	S3	Software safety assessment approach.	Approach in VERIFICATION.md.	open
3.2.2 ED-153	Plan 3.2 Planning	Partial	S3	Software safety assessment plan.	Verification plan only.	open
3.2.3 ED-153	Plan review 3.2 Planning	Non-compliant	S2	Named reviewer signs off the plan.	No independent plan review.	open
3.2.4 ED-153	Plan dissemination 3.2 Planning	Partial	S3	Distribution record.	Plans in the repo.	open
3.0.8 ED-153	Assurance rigour objective 3 SSAS	Partial	S3	State SWAL-scaled rigour in the plan.	Verification plan exists; rigour not tied to SWAL.	open
3.0.9 ED-153	Assurance rigour criteria 3 SSAS	Partial	S3	Per-objective I/R/D classification in the plan.	Verification method per requirement (T/A/I/D).	open
3.4.3 ED-153	Process assurance 3.4 V&V/PA	Non-compliant	S2	QA audit of the process.	No process audit.	open
4.2.4 ED-153	Planning 4.2 Supply	Partial	S4	Project and development plans.	Readiness report and remediation plan.	open
4.3.3 ED-153	Process implementation 4.3 Development	Partial	S4	Development plan incl. COTS scope and AI-use policy.	CONTRIBUTING and TESTING define the process.	open
4.3.11 ED-153	Requirement development management 4.3 Development	Partial	S4	Requirement change control via issues.	Requirements in git; no change control.	open
3.0.7 ED-153	Configuration management assurance 3 SSAS	Partial	S3	Tag releases; deploy a CI-built artifact.	Build stamp, lockfile, config index. Hosted demo not a clean checkout; no tags.	open
5.2.1 ED-153	Process implementation 5.2 CM	Partial	S4	CM plan.	Configuration index.	open
5.2.3 ED-153	Configuration control 5.2 CM	Partial	S4	Protected main with reviews.	Branch workflow; main unprotected.	open
5.2.x ED-153	Further CM objectives (placeholder) 5.2 CM	Partial	S4	Status accounting and release records.	Results log.	open

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
5.4.13 ED-153	Verification of retrieval and release process 5.4 Verification	Partial	S4	Tagged, reproducible releases.	Clean-clone rebuild verified; no tags.	open
3.0.12 ED-153	Software modifications 3 SSAS	Partial	S3	Per-change impact analysis template.	Branch, clean-clone verify, merge, deploy. No safety impact analysis.	open
3.0.15 ED-153	All on-line aspects of SW operational changes 3 SSAS	Partial	S3	Documented and tested rollback.	Gated deploy procedure; rollback untested.	open
4.4.1 ED-153	Process implementation 4.4 Operation	Partial	S4	Operations plan.	Demo runbook.	open
4.5.1 ED-153	Process implementation 4.5 Maintenance	Partial	S4	Maintenance plan.	Changelog and change process.	open
4.5.2 ED-153	SWAL allocation confirmation 4.5 Maintenance	Non-compliant	S3	Impact analysis per release.	No SWAL re-check per change.	open
4.5.3 ED-153	SWAL satisfaction 4.5 Maintenance	Non-compliant	S3	Release evidence pack.	No per-change SWAL evidence.	open
5.3.x ED-153	Quality assurance (placeholder) 5.3 QA	Non-compliant	S3	QA role, plan and audits.	No QA role or audits.	open
5.8.x ED-153	Problem resolution (placeholder) 5.8 Problem resolution	Partial	S4	Log all known limitations as issues.	Issue template exists; no reports raised.	open
6.1.x ED-153	Management (6.1.5 cited) 6 Organisational	Partial	S4	Project management records.	Plans and tasks tracked informally.	open
6.2.x ED-153	Infrastructure (placeholder) 6 Organisational	Partial	S4	Infrastructure definition.	CI and hosting defined in the config index.	open
6.3.x ED-153	Improvement (6.3.2 cited) 6 Organisational	Partial	S4	Improvement records.	Lessons fed back into gates.	open
6.4.x ED-153	Training (placeholder) 6 Organisational	Non-compliant	S3	Training and competence records.	No competence records.	open
7.x DE ED-153	Development environment 7 Additional ANS	Partial	S4	Environment under CM, incl. AI agent configuration.	Lockfile; Node pinned in engines.	open

W9 Requirements, verification and validation

Wave 3 · 18 findings

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
3.0.2 ED-153	Requirements correctness and completeness 3 SSAS	Partial	S3	Review and approve the SRS; derive safety requirements.	GEV-SRS 0.1 (56 requirements), reverse-engineered, unreviewed.	open
3.0.3 ED-153	Requirements traceability assurance 3 SSAS	Partial	S3	Trace or trim the inherited layer engine.	56/56 requirements traced; trace-check in CI. Inherited engine not traced.	open
3.0.4 ED-153	Unintended functions 3 SSAS	Partial	S3	Deactivated-code review; remove /api/route.	0 orphan files (import-graph gate). /api/route reachable but unused.	open
3.0.14 ED-153	Independence (of components) 3 SSAS	Partial	S3	Isolation argument per first-level component.	Import-direction and package-boundary checks enforce structure.	open
4.3.1 ED-153	System requirements analysis 4.3 Development	Partial	S4	System requirements spec.	System scope in README/ARCHITECTURE.	open
4.3.4 ED-153	SW requirements analysis 4.3 Development	Partial	S4	Reviewed SRS covering the layer engine.	SRS 0.1, 56 requirements.	open
4.3.7 ED-153	SW integration 4.3 Development	Partial	S4	Integration test plan per component boundary.	CI build and test integrate all modules.	open
4.3.14 ED-153	Rationale for design choices (esp. real-time) 4.3 Development	Partial	S4	Design decision records.	Rationale in code comments and docs.	open
3.4.2 ED-153	Verification 3.4 V&V/PA	Partial	S3	Verify safety requirements.	Functional requirements verified by tests.	open
5.4.3 ED-153	Verification of software requirements 5.4 Verification	Partial	S4	Independent review of functional tests.	Requirement-based tests pass; independence contested at SWAL 3.	open
5.4.4 ED-153	Integration verification 5.4 Verification	Partial	S4	Interface tests for every component boundary.	Server proxy and fallback tests.	open
5.4.5 ED-153	Verification of SW architectural design 5.4 Verification	Partial	S4	Architecture review record.	Boundaries verified automatically.	open
5.4.9 ED-153	Data verification 5.4 Verification	Partial	S4	Feed integrity and outage tests.	FEED/SIMULATED tagging and injection tests; no feed integrity checks.	open
5.4.12 ED-153	Verification of verification process results 5.4 Verification	Partial	S4	Verification report.	Results log per commit.	open
3.4.1 ED-153	Validation 3.4 V&V/PA	Non-compliant	S2	Validate GEV-SAF with users.	Safety requirements not validated (none exist).	open

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
5.5.x ED-153	Validation (placeholder) 5.5 Validation	Non-compliant	S3	Validation with operational users.	No operational validation.	open
5.6.x ED-153	Joint review (placeholder) 5.6 Joint review	Non-compliant	S3	Review minutes with the customer.	No joint reviews.	open
4.4.3 ED-153	User support 4.4 Operation	Partial	S4	User guide and support route.	README, runbook.	open

W10 Safety case and release evidence

Wave 4 · 7 findings

ID	REQUIREMENT	BASELINE	SEVERITY	FIX	BASELINE EVIDENCE	CLOSED BY
3.0.10 ED-153	SWAL assurance 3 SSAS	Non-compliant	S2	Direct and back-up arguments in the SSAR.	No safety argument.	open
3.0.17 ED-153	Argument that SWAL objectives are met 3 SSAS	Non-compliant	S2	GSN-style argument in the SSAR.	No argument that SWAL objectives are met.	open
3.4.4 ED-153	Documented outputs as proof requirements are met 3.4 V&V/PA	Partial	S3	Safety assessment report.	Gap analysis and readiness report.	open
3.5.1 ED-153	Document process results 3.5 Completion	Partial	S4	Assessment report per release.	Results log per commit.	open
3.5.3 ED-153	Documentation dissemination 3.5 Completion	Partial	S4	Distribution record.	Docs in the repo and project.	open
5.7.2–5.7.4 ED-153	Audit (numbers only) 5.7 Audit	Partial	S4	Independent audit.	Self-assessment only.	open
8 ED-153	Safety Folder 8 Safety Folder	Partial	S3	Software Safety Folder / SSAR.	Folder index in the gap analysis.	open

How CrossCheck rates and decides

Status scale

Compliant	Fully met, with evidence.
Largely	Met with minor, non-blocking gaps.
Partial	Some evidence; material gaps remain.
Non-compliant	Not met, or no evidence.

Weighted readiness: Compliant 1, Largely 0.75, Partial 0.4, Non-compliant 0, averaged over rated items in scope.

Severity

Safety findings S1–S4 after EASA AMC 20-189 (S1–S2 block a claim). Security findings Critical, High, Medium, Low (Critical and High block a level).

Verdict rules

ED-153 SWAL claim supported only if every required objective is at least Largely, no S1–S2 finding is open, and independence is evidenced where the SWAL requires it.

ASVS level met only if every requirement at or below the level passes or largely passes, and no Critical or High finding is open.

Evidence rules

The ledger (`docs/evidence/ledger.json`) is the single source of truth. History is append-only. Every status change cites a commit, test, document or review record.

Sources

ED-153 objectives reconstructed from EUROCONTROL *Recommendations for ANS Software* (2005), the Czech CAA ED-153 guide and EASA AMC/GM to Regulation (EU) 2017/373; confirm against ED-153 before any formal claim. OWASP ASVS 5.0.0, © OWASP Foundation, CC BY-SA 4.0.